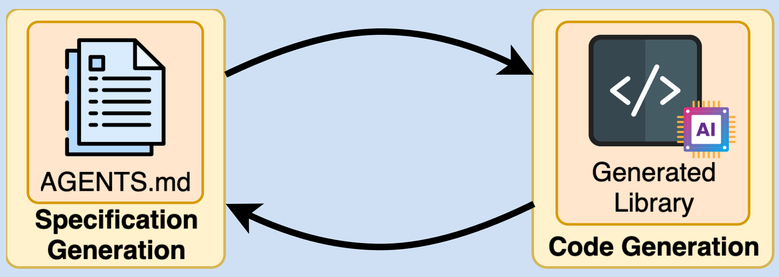
Which Context Matters? An Empirical Study of Agentic Specification Contexts Across Five Python Libraries

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1 Introduction



- AI coding agents use **context files** (e.g. AGENTS.md) to understand project knowledge and developer intent [1].
- Detailed context can improve AI-generated code, but it is unclear **which types of context are most useful** [2].
- Our previous research analyzed the **cost of creating agentic specifications** using different context types.
- This research evaluates how different context types affect **code generation**. (Full work under review)

2 Dataset & Tools

- *Python trivial libraries* [3] as the dataset.
- Experiment start date: **2026/04/26**
- **Metrics**: Lines of File **(LoF)**, File Count **(FL)**, Cyclomatic Complexity **(CC)**, Function and Methods Count **(FC)**.

Selected Trivial Library Characteristic

Library	Size	Number of Contributors
areyoustillthere	43.03 KB	1
audiodiff	2,570.55 KB	1
autolink	49.53 KB	2
bitmap	54.55 KB	7
compactor	228.87 KB	3

LLMs

Gemma-4

GPT-5.4

3 Agentic Context

Using the agentic contexts from previous study [4], we selected **five representative agentic contexts** in this research and used all possible 31 combinations:

AGENTS.md

RO — Repository Overview
- Purpose and functionality of the repository....
- High-level explanation of its functions....
- Dependencies and prerequisites....

FD — Function Description
- Function name and purpose....
- Parameters and return value....
- Side effects and errors....

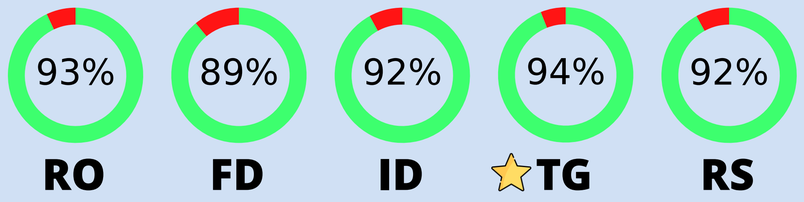
ID — Implementation Details
- Algorithm/Approach: The overall implementation strategy....
- Key Logic: Important code sections or logic requiring explanation....
- Design Patterns: Patterns or programming paradigms used....

TG — Testing
- Test Cases: Methods and cases for verifying functionality....
- Usage Scenarios: Examples of how the functions are applied....

RS — Reconstruction Steps
- Repository Setup: Steps for recreating the repository from scratch....
- Environment: Required dependencies, configuration, and setup....

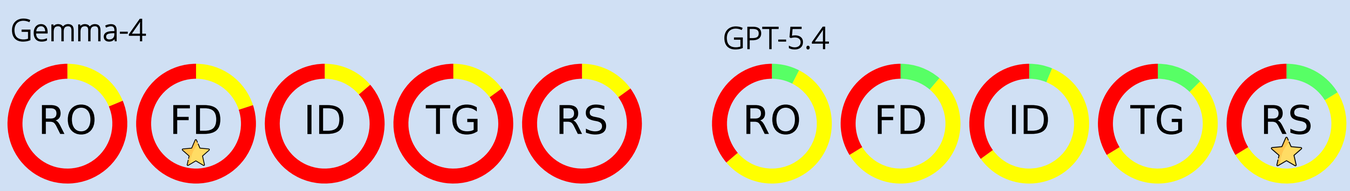
4 Key Results

RQ1: Which context exhibits the highest compilability?



- Compilability was *largely insensitive* to the types of agentic specification contexts.
- *Testing* context exhibits the highest compilability across all dataset.

RQ2: Which context exhibits the highest testability (success, partial, failed)?



- *Model choice* influences testability more than context type
- The *Function Description* context exhibits the highest testability rate for Gemma-4 (20.00% partial rate).
- The *Reconstruction Steps* context exhibits the highest testability rate for GPT-5.4 (16.25% success rate).

Model	Context	Passed (%)	Partial (%)	Failed (%)
Gemma4	RO	0.00	18.75	81.25
	FD	0.00	**20.00**	80.00
	ID	0.00	13.75	86.25
	TG	0.00	15.00	85.00
	RS	0.00	15.00	85.00
GPT-5.4	RO	7.50	56.25	36.25
	FD	11.25	55.00	33.75
	ID	6.25	58.75	35.00
	TG	12.50	53.75	33.75
	RS	**16.25**	50.00	33.75

RQ3: Which context has the most similar structural metrics to Human code?

Metric	Library	Gemma-4		GPT-5.4	
		Contexts	Δ	Contexts	Δ
LoF	areyoustillthere	ID	-76.56%	ID-**RS**	-16.44%
	audiodiff	RO-**RS**	-98.50%	TG	-76.65%
	autolink	ID-**RS**	-74.52%	RO-ID-**RS**	-66.90%
	bitmap	RO-FD-ID	-74.32%	**RS**	-70.74%
	compactor	RO-TG-**RS**	-90.52%	FD-TG	-67.17%

*Full metrics are shown in the paper (under review)

- Types of *context affect the structural metrics* of AI-generated code.
- *Reconstruction Steps* context produced the most similar Lines of File and total Cyclomatic Complexity metrics.
- Together with the *Testing* context, they produced the most similar File Count and Function Count metrics.

5 Summary

More Context ≠ More Tokens
Our findings in previous research shows that the number of contexts provided to generate agentic specifications does not align with token cost.

More Context ≠ Higher Correctness
Findings in RQ1 and RQ2 show that context type and length has limited impact on compilability and testability, so including all contexts in agentic specification files is unnecessary.

Different contexts affect different aspects of code quality.
Based on the results of RQ3, some context types affect specific structural metrics of AI-generated code. Therefore, AI agents should prioritize task-relevant contexts rather than using all contexts.

6 References

[1] J. L. Lulla, S. Mohsenimofidi, M. Galster, J. M. Zhang, S. Baltes, and C. Treude, "On the Impact of AGENTS.md Files on the Efficiency of AI Coding Agents," *arXiv preprint arXiv:2601.20400*, 2026.
[2] S. Suri et al., "Software Engineering Using Autonomous Agents: Are We There Yet?," in *Proc. 39th IEEE/ACM Int. Conf. Automated Software Engineering (ASE)*, 2024, pp. 1855–1857.
[3] Abdalkareem et al., "Why Do Developers Use Trivial Packages?," *ESEC/FSE*, 2017, pp. 385–395.
[4] Chatlatanagulchai et al., "Agent READMEs: An Empirical Study of Context Files for Agentic Coding," *arXiv*, 2025.